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CAPSTONE PROJECT · PROGRESS REPORT

FinNews Tracker

Collecting and preparing Vietnamese financial news for company-level sentiment analysis

Team

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ONE LOVE. ONE FUTURE.

Project introduction

FinNews Tracker: follow what the news says about each listed company, day by day

Why this project

- Hundreds of company news articles appear every day across many sites — too many to read.
- One article can be good for one company and bad for another: "**VNM** grows strongly while **MSN** reports a loss".
- Goal: label every news article **per (article, company)** as positive / negative / neutral.

What it shows

- Pick a **ticker + date range + news source** → number of positive / negative / neutral articles.
- News sentiment over time (chart) and the article list with links to the original.
- Home page: **tickers with the most positive news** in the last 7 / 30 / 90 days.
- Updated automatically every day.

Benefits

- **Investors:** see the news picture of a company in seconds instead of reading every article.
- **Research:** a clean, company-linked Vietnamese financial news dataset from 2010.
- Not investment advice — a reference tool alongside prices and fundamentals.

Tools

Data collection	Python · requests · lxml
Data processing	pandas · PyArrow / Parquet · rule-based company linker (regex)
Modelling	PhoBERT · PyTorch · Hugging Face Transformers · scikit-learn · VnCoreNLP
Product (planned)	PostgreSQL · FastAPI · HTML / CSS / JS + Chart.js · Docker · cron

Project workflow

This report focuses on data collection and preparation (steps ① – ③)

FOCUS OF THIS REPORT

NEXT STEPS

① **Collection**
Crawl 5 sites, 2010 → now

→

② **Cleaning**
Normalise, merge duplicates

→

③ **Company linking**
Tickers, names, former names; equity filter

→

④ **Sentiment**
PhoBERT per (article, company)

→

⑤ **Database**
Articles, companies, labels

→

⑥ **Web app**
Ticker + date + source

↻ ⑦ **Daily automated run:** crawl new articles → clean and merge duplicates → link companies → score sentiment → write to DB → web updated the same day.

Crawling

5 news sites, company and stock news from 01/2010 to today, with full article text

News site	Sections to crawl	Period	Progress (03/11/2026)
vietstock.vn	13 channels: business results · insider trading · stocks · dividends · capital raising & M&A · listings · IPO & equitization · ETFs & funds · corporate bonds · policy · fundamental analysis · covered warrants · people	01/2010 → now	Crawling Article list: **93,294 articles** (01/2010 → 02/2017); full text: not yet
diendandoanhnghiep.vn	Business *(Doanh nghiệp)*	01/2010 → now	No crawler yet
baodautu.vn	Corporate health *(Sức khoẻ doanh nghiệp)*	01/2010 → now	No crawler yet
stockbiz.vn	Business · Market · Finance · Economy	01/2010 → now	No crawler yet
dantri.com.vn	Business *(Kinh doanh)*	01/2010 → now	No crawler yet

One schema for all sites: every crawler outputs the same columns (next slide).

Broad sections (stockbiz, dantri, diendandoanhnghiep) mix in non-company news → must go through the equity news filter.

Cross-site duplicates: several sites often cover the same event → merged before counting.

Metadata

Every crawled article is stored with 13 columns

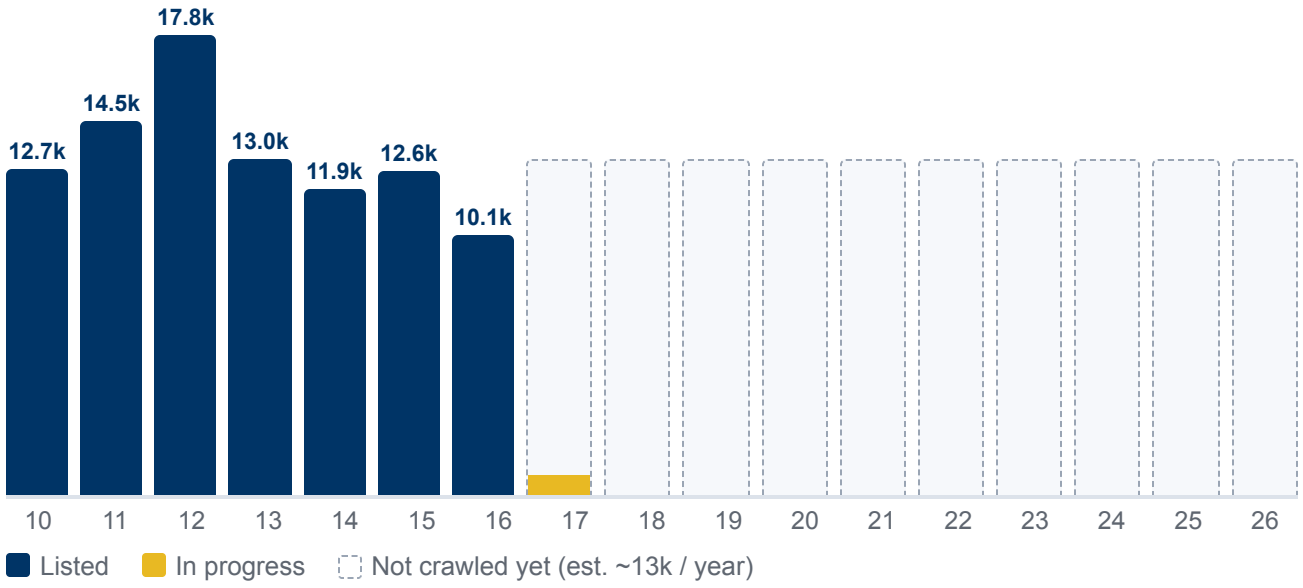
Column	Meaning	Example
`link`	Article URL — unique key, used to skip duplicates	vietstock.vn/2010/01/vna-giai-toa-co-phieu-…-739-272831.htm
`article_id`	Article ID taken from the URL	272831
`domain`	News site (source filter on the web app)	vietstock.vn
`channel_id` · `category`	Section the article was listed in	739 · Giao dịch nội bộ *(insider trading)*
`title`	Headline	VNA: Giải tỏa cổ phiếu cho cổ đông sáng lập và cổ đông đặc biệt
`description`	Lead paragraph (sapo)	*first 1–2 sentences summarising the article*
`content`	Full body text, one paragraph per line	*main input for company linking and sentiment*
`keywords`	Article tags, stored as JSON {tag: tag URL}	*{"Hòa Phát": "/tag/…"}*
`author` · `source`	Byline and cited source printed on the article	*reporter name · original publisher*
`publication_date` · `year`	Publish time (to the second) and year, used for date filters	2010-01-06 13:54:00 · 2010

Listing phase: link, article_id, channel_id, category, title, publish time. Article phase adds: full lead, content, keywords, author, source, publish time to the second.

Vietstock crawl progress

93,294 articles listed for 01/2010 → 02/2017; about 13,000 articles per year

Articles per publication year (listing phase)



By channel

Business results	32,288
Insider trading	29,017
Stocks	9,763
Dividends	6,161
Capital raising, M&A	4,435
Listings	2,988
Other channels	8,642

Article phase (downloading each page) has not started: ~1 request per article, to be split across several machines.

Daily incremental crawl

After the history is collected, the crawler only fetches new articles each day

List new articles
current month, all channels

→

Skip known links
dedupe by link

→

Download pages
new articles only

→

Save
raw_articles

→

Hand over
cleaning → linking

Resumable

Each finished (channel, month) is recorded and skipped next time; the current month is always re-listed to catch new articles.

No duplicates

Deduplicated by link; articles already downloaded are not fetched again; permanent errors (404) are logged and skipped.

Polite crawling

Minimum delay between requests, automatic retry on network errors, a log for every run.

Cleaning & duplicate merging

Normalise each article, then merge articles from different sites about the same event

Per-article cleaning

- Normalise domain (drop `https://`, `www.`) and publish date (drop "(GMT+7)")
- Remove junk characters and extra whitespace; parse keywords
- Cut trailing blocks such as "Có thể bạn quan tâm" (related articles)
- Drop advertisements (e.g. course promotions)
- Drop duplicate links

Cross-site duplicates

Sites rarely publish the same article, but often cover the same event:

vietstock	"Hòa Phát lãi hơn 8.2 ngàn tỷ trong quý 3"
CafeF	"Hòa Phát lãi 8.265 tỷ đồng, tăng 106% trong quý 3/2026"

Rule: same subject company, published ≤ 48 hours apart, same key facts (key figures or text similarity) → same `dup_group_id`, counted as **one event**; the earliest article represents the group.

Company linking & equity news filter

Linking companies first raises equity news accuracy from 79% to 96%

Former names

"CTCP Chứng khoán Beta" → **BSI** (395 former names, verified on the web)

Ticker with exchange in brackets

"CTCP Chứng khoán Sài Gòn (HNX: SHS)" → **SHS**, not SSI

Brand names

"TTTM Vincom" (Vincom mall) → **VRE**, not VIC

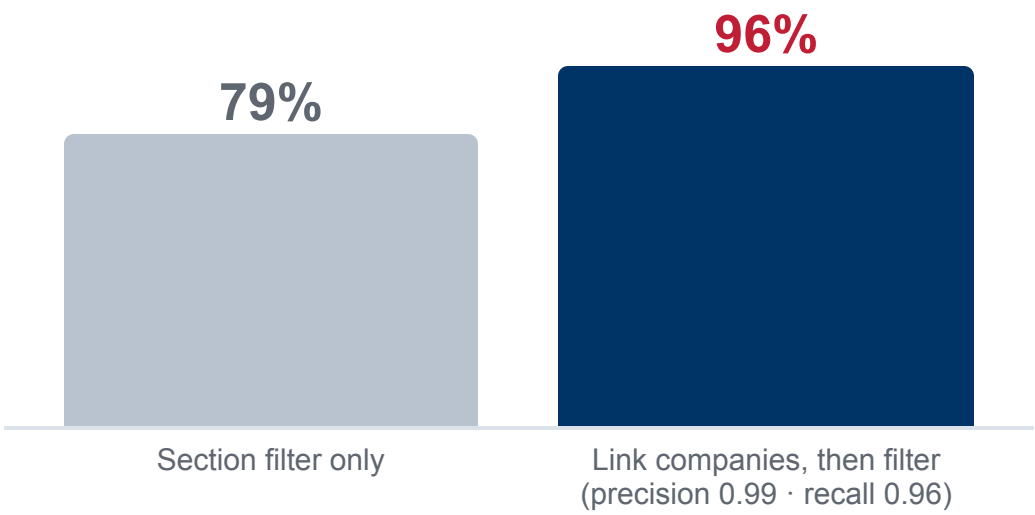
Subject vs. passing mention

Ticker in headline / lead, or mentioned ≥ 2 times → main company (similar to Tetlock et al., 2008)

1,945
tickers in the company list

85%
articles linked to ≥ 1 ticker

Equity news filter accuracy (100 hand-labelled articles)



Output data & next steps

Goal for 18/11: clean data from all 5 sites, linked to companies, ready for modelling and the web app

Output data (shared tables)

Table	Main columns
articles	article_id, link, site, section, title, lead, content, publish date, dup_group_id
companies	ticker, official name, former names, brands, exchange, status
article_company	article_id, ticker, mention count, position, is_subject, context sentences

Later steps (sentiment, web) only read these three tables.

Next two weeks (05–18/11)

- Finish the vietstock list to today; run the article phase on several machines
- Write crawlers for diendandoanhnghiep, baodautu, stockbiz, dantri
- Combine cleaning steps into one pipeline; add duplicate merging
- Link companies on the new data; measure precision / recall on ~300 articles
- Label 1,500–2,000 (article, company) pairs

Risk: crawling time

~10 years of vietstock still to list and 4 sites without a crawler; the article phase needs ~1 request per article → split across machines.